

When Does Multi-Agent Debate Help — and When Does It Backfire?

Sycophancy, dataset-dependence, and a one-line fix in cross-model LLM debate

Hwanuk Kim

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Motivation

- Lecture showed multi-agent debate as a way to RAISE accuracy (voting / debate / MDAgents).
- But is debate free? When and why might it FAIL?
 - We study the failure dynamics of debate, not just its accuracy.
 - Question: does making LLMs 'discuss' ever destroy correct answers?

Related Work & Our Contribution

- Lecture: multi-agent structures improve accuracy (voting / debate / MDAgents).
- Prior work, in tension:
 - Debate raises accuracy (Du et al., ICML'24).
 - But self-correction can DEGRADE reasoning (Huang et al., ICLR'24).
 - RLHF induces sycophancy (Sharma et al., ICLR'24).
- Our contribution:
 - (1) Diagnose sycophancy as debate's failure mode (correct answers flip to wrong).
 - (2) A prompt-only fix that reduces it (no training, no logprobs).
 - (3) A condition-dependent picture: debate helps math, hurts knowledge QA.

Method — Experimental Setup

- 3 heterogeneous models (different vendors, similar 7-8B size, non-thinking):
 - Qwen2.5-7B · Llama-3.1-8B · Mistral-7B-v0.3
- 2 datasets: GSM8K (math, 200) + HotpotQA (knowledge QA, closed-book, 200).
- 3 rounds: R0 independent -> R1 -> R2 debate (each agent sees peers' previous answers).
- Greedy decoding (temp=0): flips are caused by debate, not sampling noise.
 - 10,800 generations total (3 models x 3 rounds x 3 conditions x 400 questions).

Method — Three Conditions (Interventions)

- Same debate; only the R1/R2 prompt differs by condition (R0 is shared).
- Metrics: round accuracy, flip->wrong vs flip->correct, verbosity, source asymmetry.

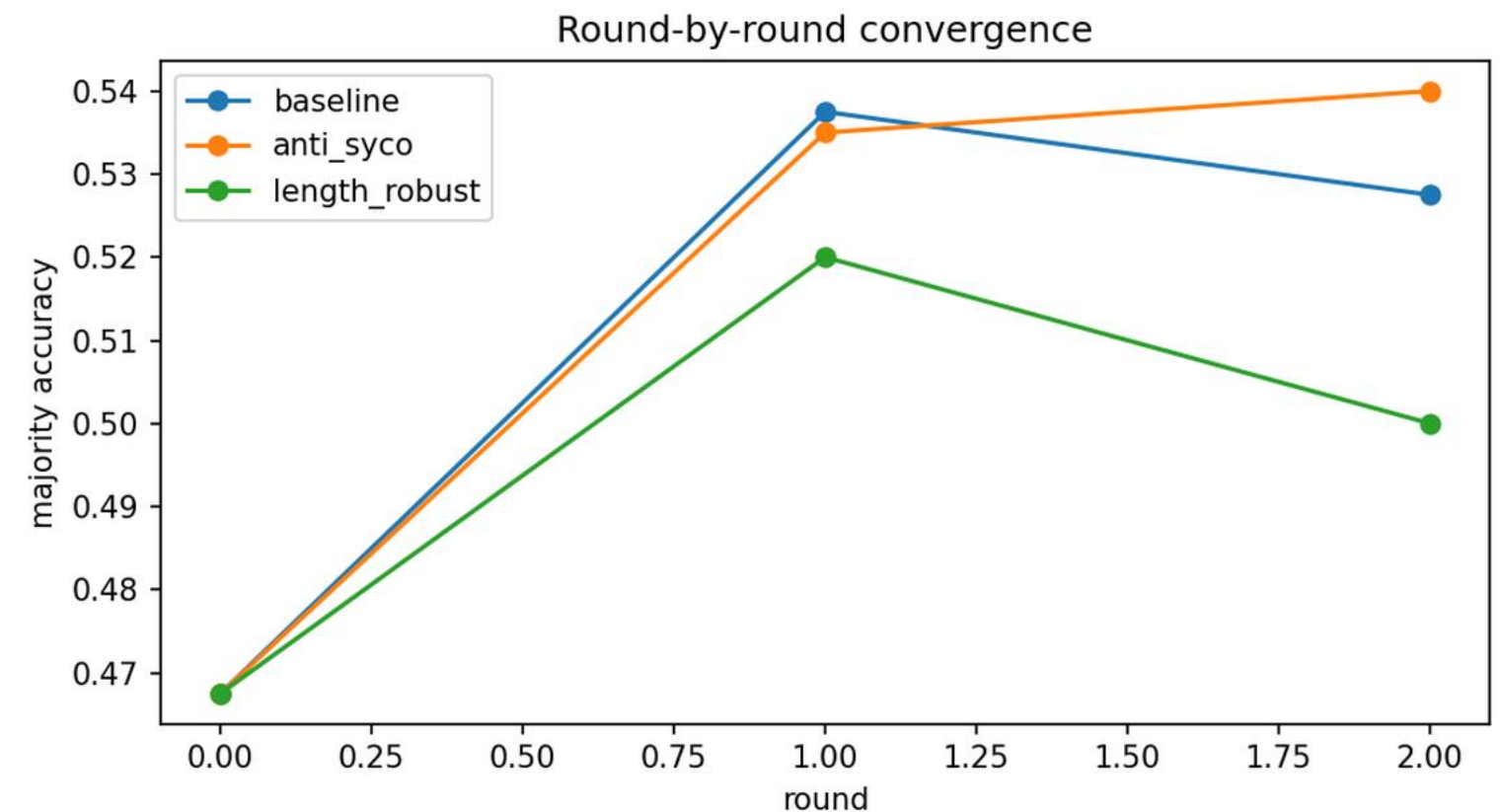
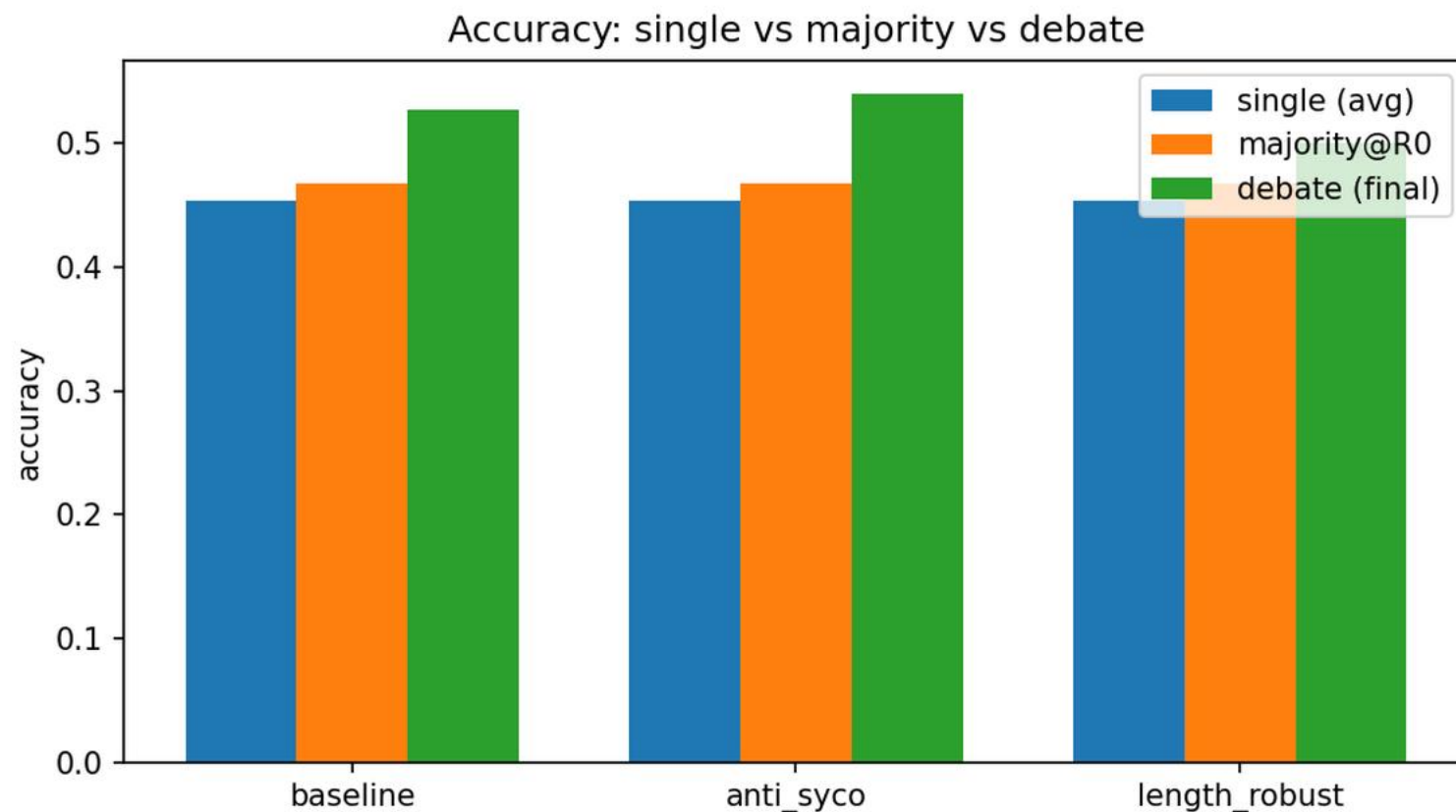
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# baseline:      peers' full reasoning is shown

# anti_syco:     one line appended to the debate prompt ->
"Do NOT change your answer unless you find a concrete,
specific error in your OWN reasoning."

# length_robust: peers shown as final answer only ->
peer = f"[Agent {name}] Final answer: {ans}"  # reasoning hidden
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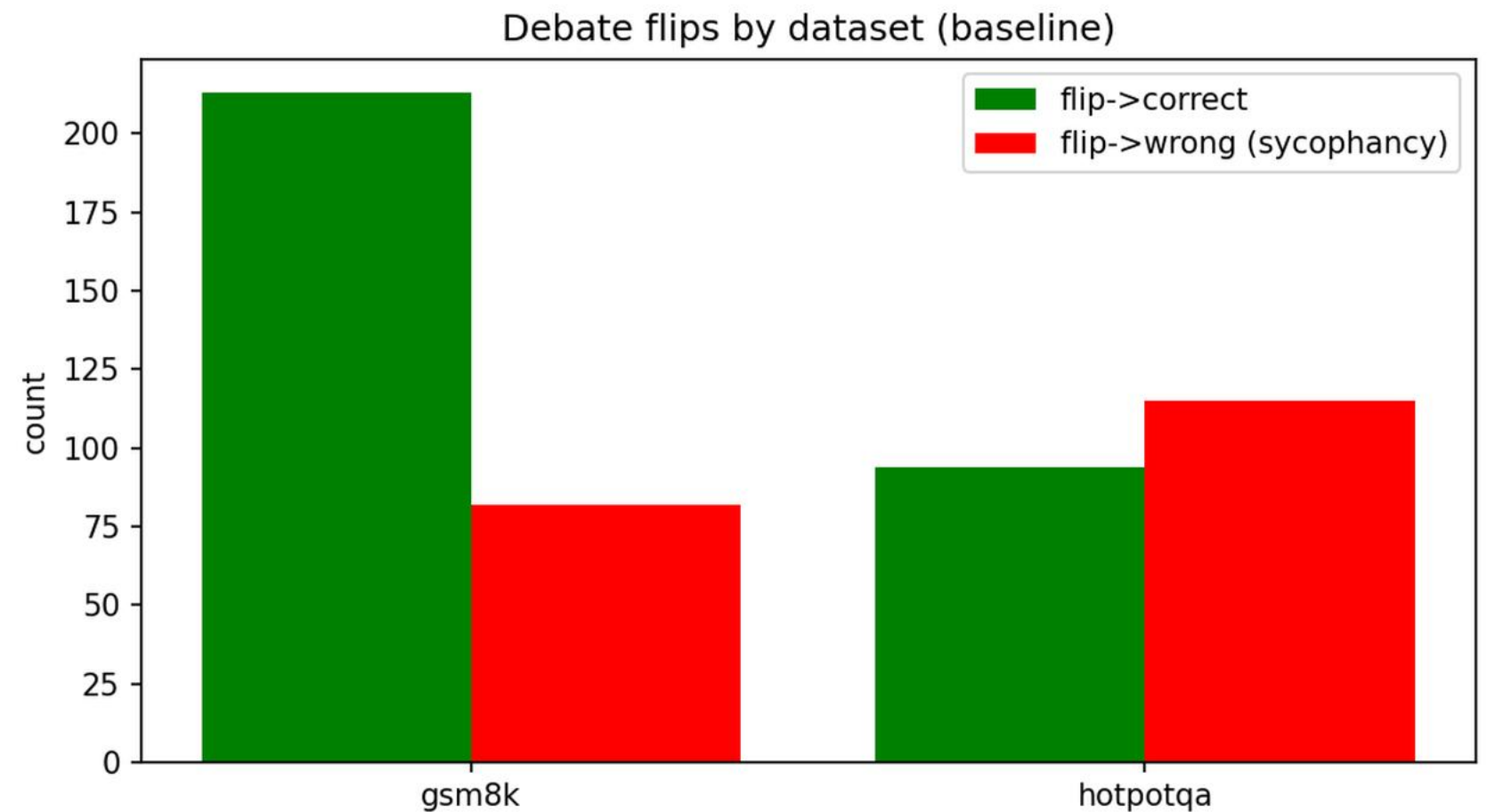
Results — Overview & Round Dynamics

- Aggregate: debate beats single answers and majority@R0.
- Most of the gain is in Round 1; Round 2 can erode it (over-debating).
- anti_syco is the only condition that resists the Round-2 drop.



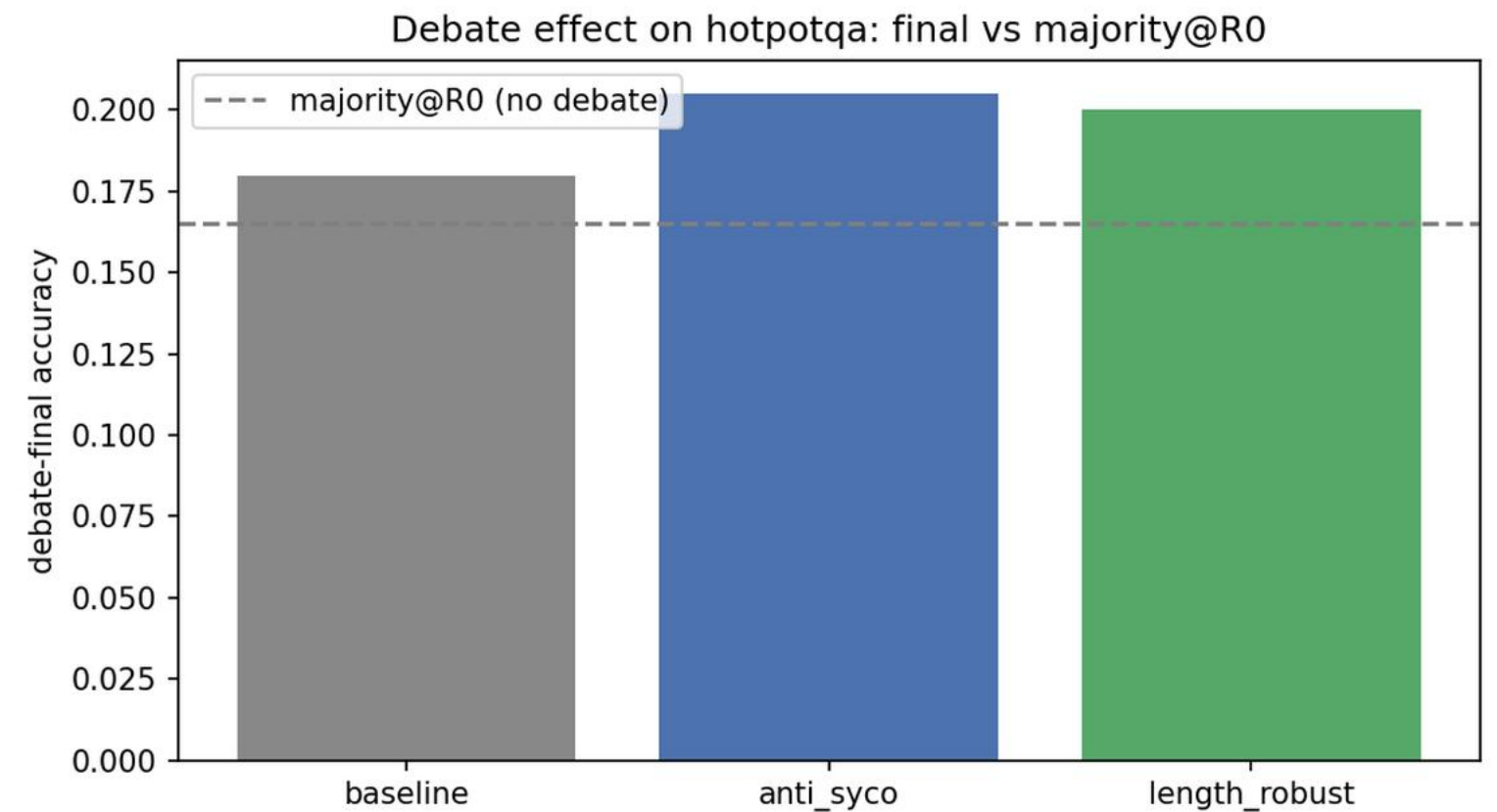
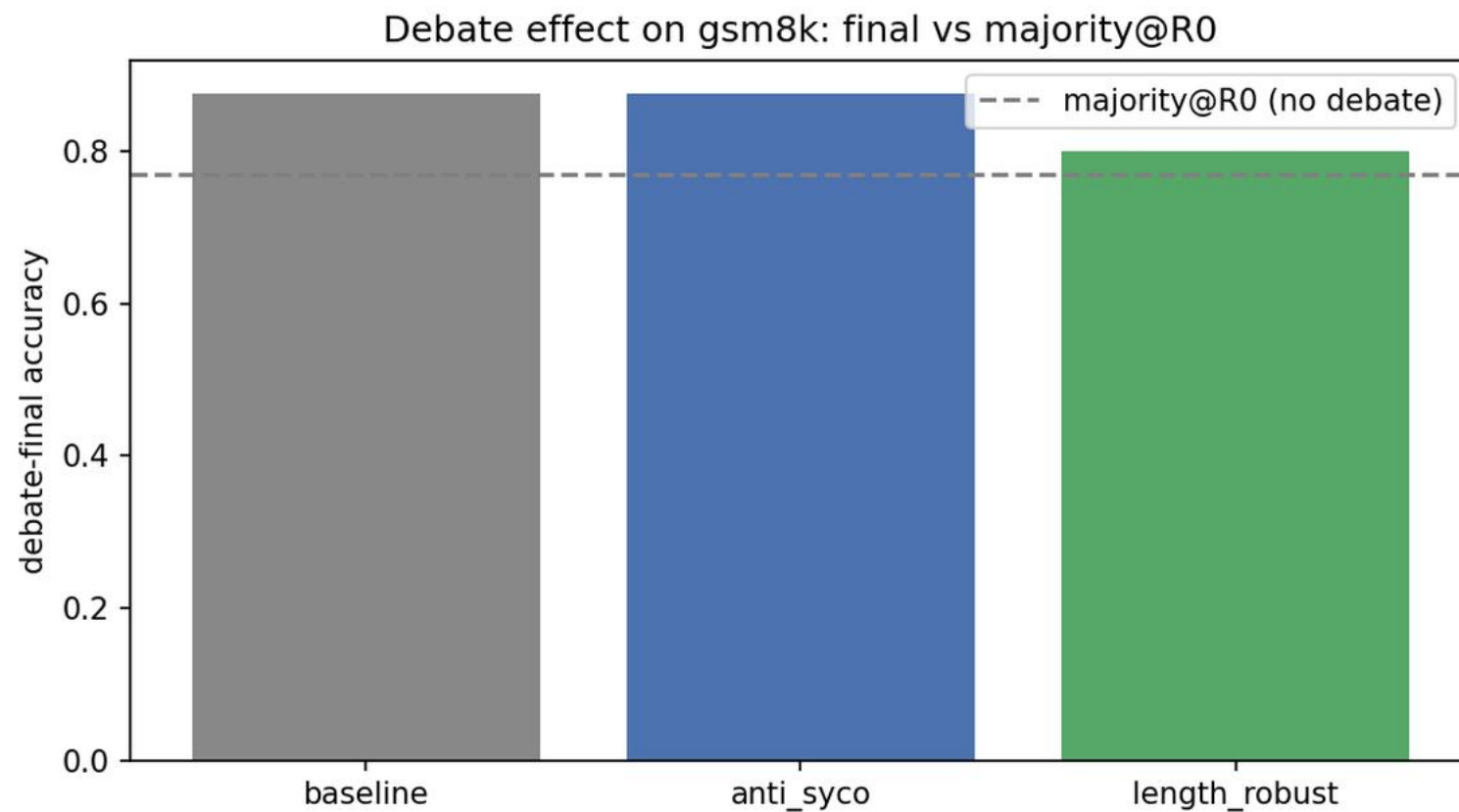
Key Finding — Debate Helps Math, Hurts Knowledge QA

- GSM8K: flip->correct 213 >> flip->wrong 82 (accuracy 0.77 -> 0.875, +10.5pts).
- HotpotQA: flip->wrong 115 > flip->correct 94 (net negative; accuracy ~flat).
 - Debate only pays off on VERIFIABLE tasks — a conditional rebuttal of the lecture's 'debate raises accuracy'.



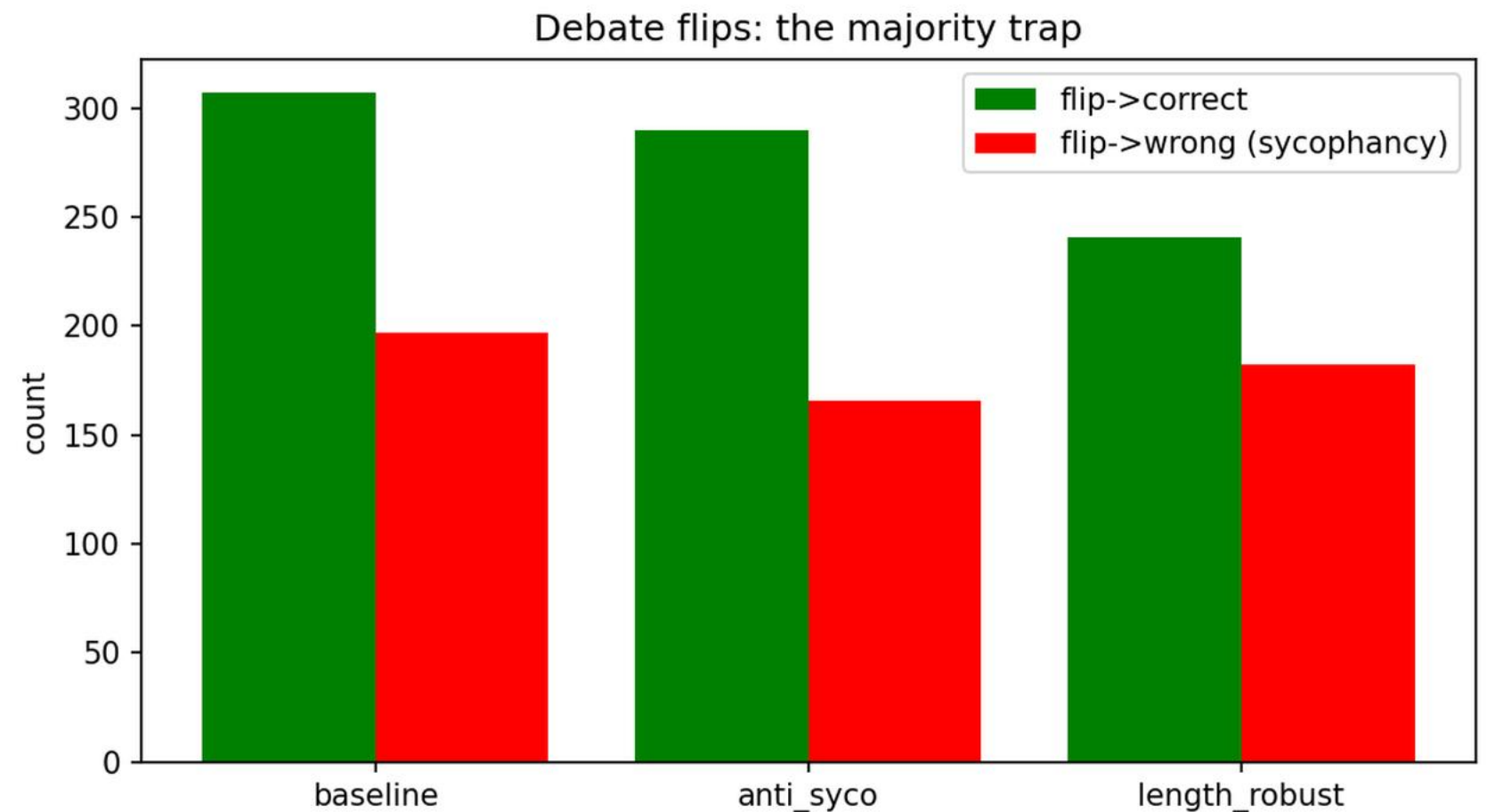
Accuracy by Dataset & Intervention

- Dashed line = majority@R0 (no debate).
- Math: clear lift; but hiding reasoning (length_robust) hurts it.
- Knowledge: small lift; both interventions beat baseline.



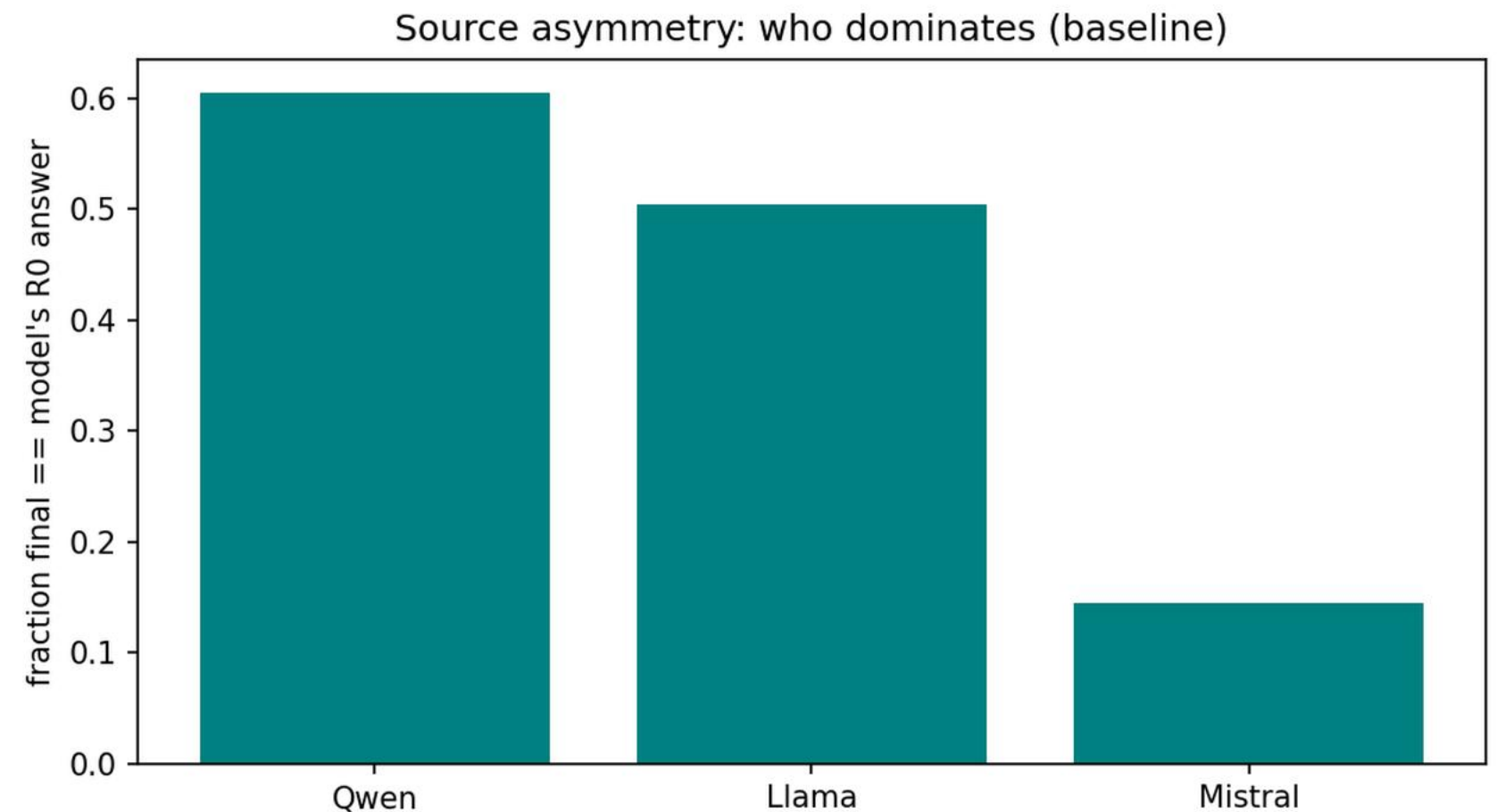
Sycophancy & the anti_syco Fix

- 'Majority trap': correct answers flip to wrong (flip->wrong, red).
- anti_syco cuts wrong-flips: 197 -> 166 overall.
 - On HotpotQA (where debate hurts most):
flip->wrong 115 -> 98, accuracy 0.18 -> 0.205.



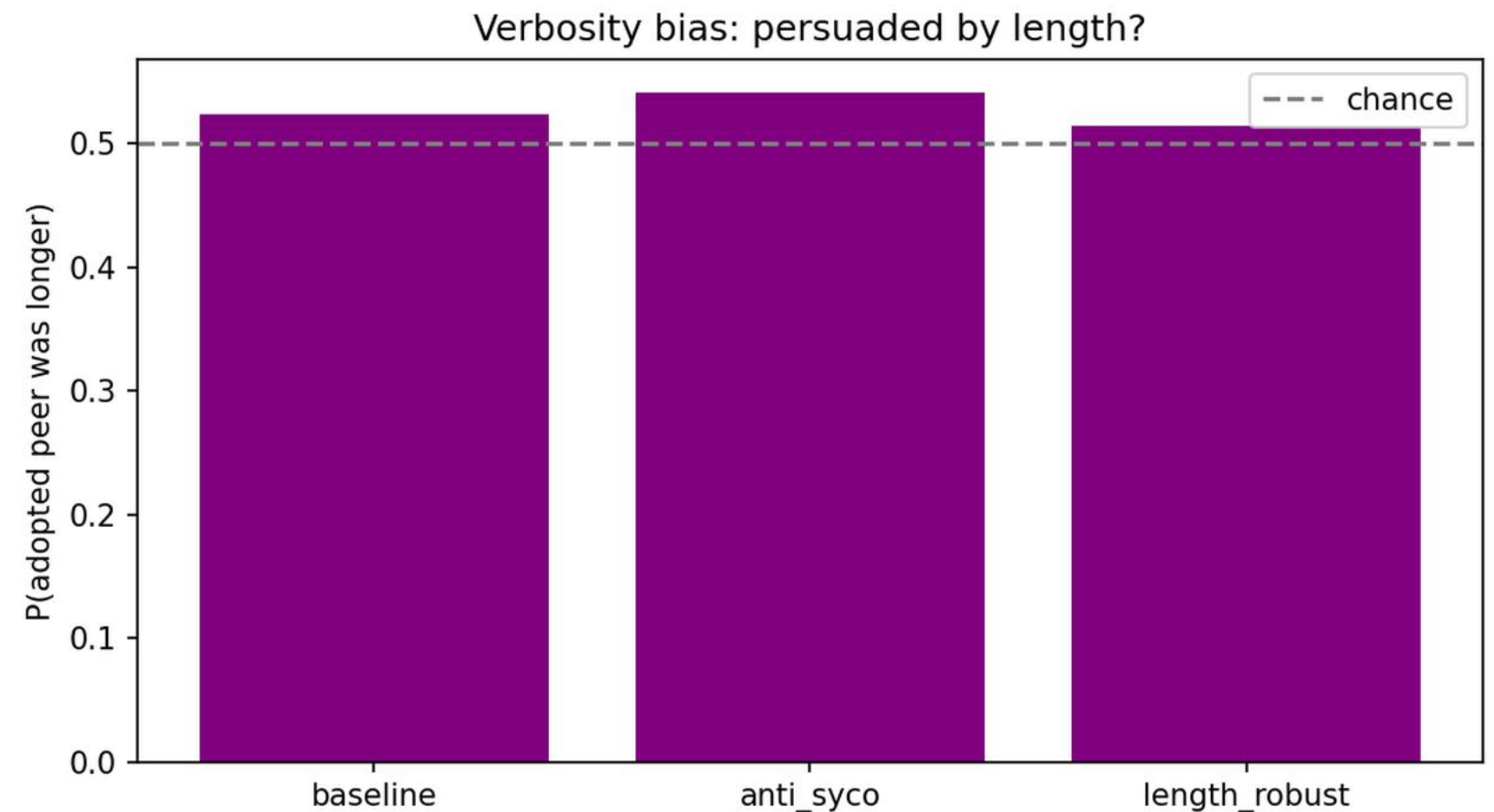
Source Asymmetry — Skill, Not Length, Dominates

- How often the final consensus matches each model's R0 answer:
 - Qwen $\sim 0.60 >$ Llama $\sim 0.50 >>$ Mistral ~ 0.14 .
- But average length is Llama (722) $>$ Qwen (500) $>$ Mistral (343).
 - The dominant model (Qwen) is not the longest $\rightarrow$ consensus is driven by capability, not verbosity.



Honest Negative + Open Problem

- Verbosity bias was WEAK:
 - $P(\text{adopted peer was longer}) \approx 0.52$ (chance).
 - The 'persuaded by length' hypothesis is rejected at scale.
- Open problem:
 - Why do confident-but-wrong answers win in knowledge QA?
 - Does consensus track confidence rather than correctness?
- Limitations: 3 models, 7-8B, 2 datasets; heuristic interventions.



Conclusion

- Debate is not free magic:
 - Big win on math (+10.5pts); net-negative flips on knowledge QA.
 - A one-line anti-sycophancy prompt reduces wrong-flips, most where debate hurts.
 - Consensus is dominated by model skill, not answer length.
- Contribution: diagnose -> intervene -> condition-dependent analysis.
 - Code + demo video included.

References

- Most relevant prior work:
 - Du et al. Improving Factuality & Reasoning in LMs through Multiagent Debate. ICML 2024. arXiv:2305.14325
 - Huang et al. Large Language Models Cannot Self-Correct Reasoning Yet. ICLR 2024. arXiv:2310.01798
 - Sharma et al. Towards Understanding Sycophancy in Language Models. ICLR 2024. arXiv:2310.13548
- Datasets & models:
 - GSM8K (Cobbe et al., 2021) · HotpotQA (Yang et al., EMNLP 2018)
 - Qwen2.5-7B · Llama-3.1-8B · Mistral-7B-v0.3 | Serving: vLLM